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DS4200

Final Project: White Hat/Black Hat Visualization Write-Up

Introduction:

This project was inspired by my own experience in my hometown San Francisco, California. Across the United States, San Francisco is known for its pervasive and persistent affordability problems and equally known for its homelessness problem. With this project, I hoped to explore not only what work San Francisco was doing to provide affordable housing for its citizens, but more importantly, where it was being built. Affordable housing built in commercial or industrial districts aren't attractive and often aren't convenient for people to actually live in. Whether that means they are unfeasibly far from grocery stores, public facilities like parks, or public transportation, people cannot live in places that don't support residents. This project's goal was to address the research question of: "Where is affordable housing being built in San Francisco."

The datasets were taken from data.sfgov.org (San Francisco's Open Data provided by the city and county). The first contains information about affordable housing projects being built in San Francisco. It is called the "Mayor's Office of Housing and Community Development Affordable Housing Pipeline" and is maintained by the Mayor's Office of Housing and Community Development (MOHCD) and the Office of Community Investment and Infrastructure (OCII). It contains all projects as of September 30, 2023 and was last updated May 13, 2025. The second dataset is called "Map of Supervisor Districts (2022)" and contains geospatial data on the City of San Francisco and its supervisorial districts. It reflects the latest updated district map after the 2022 redistricting.

Dataset Description:

Preprocessing steps

1. Scanning for incomplete records.
2. Adding a column to represent total affordable housing being built by neighborhood.
3. Adding a column to represent the directional region of each project.
4. Fixing an error in the geojson data of San Francisco as there was a weird point that created a blob out in the Pacific Ocean.

As I was preprocessing the data, I also was also doing EDA and looking for patterns within the data. One very interesting thing that I found was that while most neighborhoods fell into the same supervisorial district, some of them had projects from up to three different districts. This sparked my curiosity since neighborhoods are cultural boundaries whereas districts are political. This meant that some neighborhoods were being represented by multiple supervisors and some supervisors represented multiple neighborhoods. I was very interested to see what differences there were between directional region and supervisorial district and neighborhood when it came to having affordable housing being built.

The relevant variables that were checked for incomplete records and used in the visualizations are as follows:

1. Project Name (Housing Pipeline Dataset)
    a. Used to identify different projects within the map visualization
2. Supervisor District (Housing Pipeline Dataset)
    a. Used to juxtapose neighborhood in where affordable housing is being built in both visualizations
3. City Analysis Neighborhood (Housing Pipeline Dataset)
    a. Main grouping in the bar graph visualization to see where affordable housing is being built
4. MOHCD Affordable Units (Housing Pipeline Dataset)
    a. Used to determine the number of units being built in each neighborhood, being built in each district, and physically in different places within San Francisco
5. Latitude (Housing Pipeline Dataset)
    a. Used in the map visualization for locating each project
6. Longitude (Housing Pipeline Dataset)
    a. Used in the map visualization for locating each project
7. The multipolygon geometry (Map of Supervisor Districts)
    a. Used in the map visualization to create the boundaries of San Francisco and its districts

Design Rationale

Key decisions made in the white-hat visualizations:

Stacked Bar Chart

Having "Neighborhood" on the y axis and "MOHCD Affordable Housing Units" on the x axis was the first choice that I made with my visualizations. This was because I view neighborhoods as the cultural boundaries of the city and an anchor point that San Franciscans all understand. Everyone can name the neighborhood they live in (or that they perceive they live in), but fewer can name the numerical district that they live in. However, I did decide to include supervisorial districts as the color coding because, like we learned in Module 6, district lines are political in nature and determine differences in policy based on the way they are governed. They also give a defined boundary, providing an objective measure, whereas a neighborhood is defined by the feelings of belonging by residents themselves. This led to once again seeing that some neighborhoods were represented by multiple supervisors and supervisors sometimes represented multiple districts. I decided to orient the stacked bar chart horizontally and sort it by neighborhood with the most affordable housing units being built to the least in order to facilitate comparison and quickly point viewers to the answer to the research question. I also decided to represent the sum of the affordable housing that was being built in order to simplify the viewer's experience and not over-complicate the visualization. Seeing the visualization for the first time, you can easily see that affordable housing is being built in Bayview Hunters Point, the Mission, the Financial District/South Beach and South of Market in terms of neighborhoods and in districts 6 and 10 in terms of supervisorial districts. However, I felt that if I was going to include supervisorial districts, I should provide an easier way of comparison than searching for the different colored bars. So I also decided to add a supervisor district selector where the viewer could select which district to explore and see the neighborhoods that that supervisor was building affordable housing in and how much. I also decided to create a tooltip to allow for further investigation by the viewer as to not only how many units the neighborhood was building but also how many units each district was building within that neighborhood. Precise numbers allow the viewers to compare neighborhoods with extremely similar numbers of units being built, such as Visitacion Valley versus Chinatown versus Bernal Heights. All together, this visualization was created to be as unbiased as possible, without pushing a particular narrative, but rather to increase accessibility to this data and allow viewers to come to their own conclusions.

Geospatial Map

Once I had completed my first visualization, I noticed that it was not a very accessible visualization to those who do not live in San Francisco or are not readily informed on the various neighborhoods and supervisorial districts. I created a map to supplement the stacked bar chart because it was able to physically show a viewer where the different affordable housing projects were being built rather than just giving them an abstract name of a neighborhood or number corresponding to a district. In order to not overwhelm the viewer with circles, I decided to group the different projects by their directional region within San Francisco. These regions were color coded to different colors than the stacked bar chart in order to not confuse the viewer into thinking that they translate from one visualization to the other. Circles were scaled with the default Vega-lite scaling so as the number of units in a project got larger, the circles were appropriately enlarged. Another design choice that I made was to label each of the districts to maintain some continuity with the stacked bar chart. This way, viewers can see where specifically the units from the bar chart were being built. For example, if a viewer noticed that district 6 seems to be represented heavily in the stacked bar chart, they can confirm on the map and see all of the circles that represent the different projects. Because of the sheer number of neighborhoods that San Francisco has (32 in total), it was unfeasible to represent them either by color or by outline because then the visualization would become too chaotic. I opted instead for a multi-select tool sorted by directional region where a viewer can select the neighborhoods they are interested in. This also helps with maintaining continuity with the stacked bar chart because, just like the districts, a viewer could find which neighborhood they were interested in seeing in the stacked bar chart and see where they were physically being built within San Francisco. Lastly, I created a tooltip that identified each project by its name, gave the neighborhood and district, gave the number of units that the particular project was building, and also showed the total number of units being built within that neighborhood. Similarly to the stacked bar chart, this visualization was meant to help viewers interact with this data in a more accessible way and to allow them to draw their own conclusions through exploration.

Key decisions made in the black-hat visualizations

Stacked bar chart

This stacked bar chart contains elements very similar to its white-hat counterpart but has some variations to mislead the viewer. The first design choice I made was to artificially change the unit axis to represent 50 to 1600 instead of 0 to 1800. This had the minor effect of slightly exaggerating the differences between all of the neighborhoods but has the much more significant effect of misrepresenting many of the neighborhoods' unit total. Bayview Hunters Point now looks like it ends at 1600 but if you mouse over it, the tooltip will still accurately tell you that there are 1773 units being built. Meanwhile, for any neighborhood not building more than 50 units, it looks like they are building 0. And to top it off, 50 isn't that big so you can't see that the axis has been cut off on the low end unless you really look hard. The district colors have been reassigned to a red to blue color scheme and, together with the title, axes labels, and legend labels, imply some kind of moral duty or shirking. The chart is also sorted strangely in a valley shape with the red districts on the left and the blue on the right suggesting some kind of trend of more red (bad) to neutral to more blue (good). In all, this visualization is meant to mislead the viewer into thinking that the neighborhoods on the right are doing a better job than the neighborhoods on the left, even though the sorting means nothing and the colors are categorical, not quantitative.

Geospatial map

Similarly, this map contains almost identical elements to its white-hat counterparts but tries to push a narrative using misleading features. First, the circles are scaled using an exponent to all be pretty big, making a project with 4 units look pretty similar to a project with 300 units. The regions (which have been renamed to Zone Designations) are now on an increasing opacity scale of red. Together, they push the narrative that there is a crisis in San Francisco and, in particular, in the places that the stacked bar chart suggested were the best (or carrying the burden). With the higher opacities of red representing the blue on the stacked bar chart and the lower opacities of red representing the red on the stacked bar chart, it's meant to confuse viewers just enough that they give up and don't think too hard about the framing and instead focus on the alarming amount of red. The map is also shifted more on the South and East to automatically center the viewer's eyes on the "problem zone". It even cuts off part of Treasure Island as if to say that it isn't as important. With subtle naming changes such as "Affordable Housing" to "Housing Density Burden", "Region" to "Zone Designation", "Project Units" to "Alleviated Units", and "Total Neighborhood Units" to "Total Contribution by Neighborhood", this map uses scary language to point the finger at Western and Central San Francisco as not doing their fair share while suggesting that the South and East are in crisis.

Development Process

Timeline

- Created first draft of white-hat stacked bar chart as a part of Assignment 2

- Incorporated a dropdown menu into the white-hat stacked bar chart as a part of Assignment 3
- Created first draft of white-hat geospatial map as a part of Assignment 4
- Heard feedback from peers on what worked and what did not work with my proposed white-hat visualizations
- Decided that my two white-hat visualizations would not be combined but used in conjunction
- Finalized design of white-hat stacked bar chart as a part of Assignment 4
- Added selector for the white-hat geospatial map as a part of the Final Project
- Finalized white-hat geospatial map as a part of the Final Project
- Created the black-hat stacked bar chart as a part of the Final Project
- Created the black-hat geospatial map as a part of the Final Project

Tools used

Throughout my visualizations, I tried to make them as interactive as possible while keeping the quickness of providing information in mind. I used tools such as tooltips and selectors to allow for further exploration of the data, should the viewer be interested, but left a default that showed all the data in case the viewer just wanted a quick answer. I particularly like the multi-selector that I was able to create for the map as it is a great link to the stacked bar chart and allows for much more exploration than a single selector.

Major challenges

1. Whether to show affordable housing units by neighborhood or district

    I ultimately decided to put neighborhoods on the axis and encode districts as color because I believe that cultural boundaries are more significant than political ones when it comes to providing housing for fellow citizens but districts are still important as they determine what kinds of policies are being enforced.
2. The stacked bar chart was not easily understood by viewers not familiar with San Francisco

    I decided that the best way to alleviate this was to create a second part to the project with a map showing where the housing developments physically were.
3. How could I break up the sheer number of circles on the screen into something more digestible

    I decided to carve out regions of San Francisco so that the circles were not overwhelming the viewer with a single color but rather could focus on individual sections at a time.
4. Should I somehow combine the two visualizations into one

    I decided that for clarity's sake, having them separate would enhance the understandability, especially after adding many links that would help with information continuity. I also thought that it would benefit the viewer to see both at the same time so that they could track data between the two visualizations instead of having to flip from one to the other.
5. How can I represent neighborhoods in the geospatial map in a way that was digestible for a viewer

    Originally, I had them color-coded on the map but after peer review, they helped me determine that 32 colors was too chaotic for the visualization. I ultimately decided to create a multi-selector so that viewers can choose which neighborhoods to see at a time.

6. How could I use the black-hat visualizations to subtly mislead the viewer

   I decided to create a theme of burden, with some neighborhoods "carrying more weight" than others. It happened that I was able to sort them in a very misleading way in the stacked bar chart and it transitioned well to the map where I was able to misrepresent affordable housing units being built as areas most burdened.
7. Whether to mislabel the districts on the map

   I had originally thought it might be interesting to mislabel the districts on the map to match up with the narrative in the stacked bar chart but ultimately decided against it as it took the visualization from being misleading to lying.

Feedback Incorporation

Peer feedback was very valuable in the revisions of my project and I appreciate all of my group mates for giving me critiques. One of the biggest suggestions was to change the color encoding on the map from neighborhoods as they found it confusing and overwhelming. That was a great idea and I completely agreed with them so I changed it from neighborhoods to regions. Another thing that multiple people suggested was to add the outlines of political boundaries to the map as it only had the outline of San Francisco. I had originally drawn the boundary myself but, luckily, I was able to find a geojson file that worked with vega-lite with minimal need for pre-processing. One suggestion that was super interesting to me was if selecting a particular district or neighborhood on the bar chart could highlight that area on the map. While I invested some time into figuring out if this would work, ultimately it proved too difficult. But I did take that idea into mind and created many more links between the two visualizations to maintain continuity of the data. Lastly, peer feedback was the reason I ultimately decided that I would keep both visualizations and not combine them as originally, I was thinking that I would combine them through some kind of animation. A peer suggested that both of them, side-by-side, was an option and I thought it was a good suggestion. A subtitle of the time range was also included in each of the visualizations after suggestion by a peer.

Final Reflections

I am overall super happy with the way I was able to represent the data through my two visualizations. It took a lot of hard work and a lot of ideation but I think that any viewer who is interested in where affordable housing is being built in San Francisco will have their questions answered by these visualizations. Throughout building these visualizations, I learned many lessons. I learned that sometimes using just one visualization is not enough to fully answer the question that you are asking. I learned how colors can evoke emotion and how they can suggest links that you may be un-intending to suggest. I learned how the use of titles, axes labels, and legends can also push narratives if you are not careful. Most of all, I learned that the same dataset can tell opposing stories, depending on how you represent it. Data can be technically accurate but ethically problematic when you use particular techniques to push an agenda. But on the other side of the coin, if represented using ethical techniques, you can impart interesting information on the viewer and allow them to create their own conclusions based on the data in front of them rather than force-feeding them a narrative that you created yourself.

Next time, I might look into different technologies to create these visualizations. Vega-lite has a lot of nice built-in defaults but I would like to explore linking the visualizations like one of my peers suggested. I would also look into having an updating number at the bottom of the visualizations to represent how many units are being built in the subset of San Francisco that the viewer is inspecting as it could be cool to see the difference in totals of the different areas.

Acknowledgements

Mayor's Office of Housing and Community Development Affordable Housing Pipeline | DATASF. (2025, May 13). https://data.sfgov.org/Housing-and-Buildings/Mayor-s-Office-of-Housing-and-Community-Developmen/aaxw-2cb8/about_data

Map of Supervisor Districts (2022) | DataSF. (n.d.). https://data.sfgov.org/Geographic-Locations-and-Boundaries/Map-of-Supervisor-Districts-2022-/63dt-nj2t